Lightweight Trap-Based Free-Rider Screening with Fresh-Certificate Memory in Federated Learning

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***Abstract*—Free-rider screening must distinguish fabricated updates from heterogeneous honest learning without relying on attacker identities or client-reported work. We describe a server-observable challenge protocol combining same-trap robust norm and tensor-profile measurements with fresh separation certificates. Overlapping low-norm cohorts cannot authorize removal through concentration alone. Statistical certificates persist across ambiguous cycles, but confirmation requires a fresh certificate of the same component on both traps. Two consecutive clean cycles clear memory; exact sendback has a separate consecutive-cycle path. This revision provides reproducible templates for sixteen CIFAR-10 executions: IID and Dirichlet non-IID partitions, FR1–FR4, 30% and 40% free riders, seed 42. Numerical results and curves are withheld until every execution completes and its logs validate. The protocol is an empirical geometric heuristic, not a calibrated false-positive bound or proof of training. A development seed cannot establish generalization to new partitions or adaptive adversaries.**

***Index Terms*—federated learning, free riders, robust statistics, challenge models, non-IID data**

I. Introduction and Related Work

Federated learning aggregates locally computed contributions while retaining training data at clients. Model averaging is a foundational example of this approach [1]. Participants that consume the shared model without performing expected training create a free-rider problem. Constructed updates can be nonzero and history-dependent, making equality testing alone insufficient. Prior work studies fabricated contributions and defenses [2], [3].

Heterogeneous honest data complicates attribution: persistent low update norms can reflect legitimate learning rather than misconduct. Conversely, a fabricated response may be anomalous repeatedly without satisfying adjacent-cycle confirmation. We therefore distinguish nomination, fresh separation evidence, temporary review, and permanent removal. This revision describes v15.4, not the older v8 anchor/penalty method. Its contributions are a strict norm gate, component-specific certificate memory, bounded review independent of memory, and separate accounting of suspicion, removal, exclusion and utility. All scheduled outcomes must be reported, not only successful attacks or seeds.

II. Threat Model and Simulator Attacks

The simulator uses 100 attributable clients and full participation among clients not permanently removed. The detector uses received/returned states, public IDs and its own schedule. It does not consume attacker labels, attack type, private loss, sample count, telemetry or an expected removal count. The simulator knows attack configuration to fabricate responses; evaluation labels are appended after decisions. This API boundary does not imply that the whole simulation application is unaware of ground truth.

Attackers receive no trap-status flag. Their evaluated implementations use only their own received-model histories and local randomness. This is not a cryptographic unpredictability claim against a seed-aware adversary. Collusion, deliberate evasion, partial/on-off training, identity changes, dropouts, secure aggregation and populations without an honest majority are outside the study.

The four behaviors are simulator settings, not detection signatures:

- **FR1: current sendback.** Return the currently received model unchanged, not the previous-model playback described in the v8 paper.
- **FR2: bounded random update.** Add independent uniform noise in $[-10^{-3}, 10^{-3}]$ to each floating coordinate of the received model.
- **FR3: received-history averaging.** Return the mean of up to five received models including the current one. Traps enter the same history.
- **FR4: received-trajectory imitation.** Average up to five successive received-model differences and add Gaussian noise whose norm is 10% of the predicted norm. Cold start uses Gaussian scale 10^{-4}; implementation norm normalization has numerical floor 10^{-6}.

The detector does not reconstruct these history windows. Different windows remain a required follow-up experiment rather than an established robustness claim.

III. Proposed Methodology

A. Warm-Up and Challenge Coverage

Rounds 1–10 send the ordinary global model. Sendbacks are observed, but warm-up cannot remove clients. At each

dense-cycle start, create two independent traps around that cycle's frozen global model. For floating state tensor k,

$$\tau_{a,k} = w_k + \gamma \max(\mathrm{sd}(w_k), \epsilon_t)\xi_{a,k}, \quad \xi_{a,k} \sim \mathcal{N}(0, I), \quad (1)$$

where $a \in \{A, B\}$, $\gamma = 0.01$, $\epsilon_t = 10^{-4}$. Traps A/B are independent, not complementary plus/minus perturbations.

Ten shuffled groups give each client one response to each trap during rounds 11–20. Later cycles form up to ten nonempty groups from monitored and ordinary pools separately, with proportional nearest rounding and half ties downward. Monitored clients receive alternating traps in all ten rounds; ordinary clients receive each trap once. A single response position per client per trap is selected before scores are observed. Extra responses cannot become additional confirmation strikes in the cycle. Fresh traps and grouping are used each cycle.

Only bounded-review clients automatically receive dense monitoring. A weak C label does not impose indefinite probing or quarantine. When review/probation are empty, an all-client single-trap audit occurs every ten rounds. It can initiate bounded follow-up, but cannot confirm removal or clear certificate memory.

B. Same-Trap Norm and Tensor Profile

Define $\Delta_i = w_i^{\mathrm{ret}} - w_i^{\mathrm{send}}$ and $r_i = \|\Delta_i\|_2$ over floating tensors. Profile coordinate $p_{i,k} = \|\Delta_{i,k}\|_2 / \sum_j \|\Delta_{i,j}\|_2$ describes tensor-wise norm allocation. A denominator below 10^{-6} yields a zero profile; that convention is not the exact-sendback criterion.

Each selected response is compared with all other available active responses to the same trap, without trusted-R or ground-truth filtering. For peer norm median m_i and MAD d_i,

$$z_i^n = \frac{r_i - m_i}{d_i / \Phi^{-1}(0.75)}. \quad (2)$$

Zero MAD produces an undefined score, not an arbitrary MAD floor. Normal consistency is not a Gaussian-update assumption. Profile distance is Euclidean distance from the peer coordinate-wise median profile. Its robust score uses peer distances about that same reference. Cosine is not collected or used.

C. Nomination Is Not Removal Evidence

Existing diagnostic screening uses $D_i = z_i^p - z_i^n$, its A/B values and their average. Each view uses $\lceil \sqrt{N} \rceil$ histogram bins, an Otsu between-class-variance boundary, and 1,000 whole-client bootstrap resamples with a fixed reproducibility seed. All three values above the upper 99% boundary-stability bands nominate suspicion; all below the lower bands are unflagged, otherwise C. An unsupported majority split falls back to norm signs about same-trap medians. These nominations cannot alone remove clients. The 99% band is not a 1% false-removal guarantee.

D. Fresh Component Certificates

Let C be the nominated cohort and P its complement for one trap. Require $\lceil N/\lceil \sqrt{N} \rceil \rceil \le |C| < |P|$. With cohort spread $I_C = Q_{.75}(C) - Q_{.25}(C)$, the strict low-norm gate requires

$$Q_{.25}(P) - Q_{.75}(C) > I_C, \quad (3)$$

$$\min(P) - Q_{.75}(C) > I_C. \quad (4)$$

An individual must also lie below the peer median. Concentration alone cannot authorize norm removal. This separates a robust core, not necessarily every member: the upper quartile is not the cohort maximum.

High raw profile distances retain the v15.1 gate. Let I_P be peer IQR and $g = Q_{.25}(C) - Q_{.75}(P)$. Require $g > I_C$, $g > 0$, and either

$$Q_{.25}(C) - \max(P) > I_C \quad \text{or} \quad \frac{I_C}{|C| - 1} < \frac{I_P}{|P| - 1}. \quad (5)$$

The individual must exceed the peer median. The same component must certify on both selected traps. Invalid/missing measurements cannot certify. Diagonal scores remain diagnostic because they already contain norm evidence.

E. Memory, Clearance and Exact Sendback

Retain $M_{i,j}$ separately for norm and profile components. A fresh full A/B certificate increments its own count once per distinct completed cycle. Ambiguous C cycles pause without incrementing or erasing counts. Two consecutive complete R cycles clear them. Statistical removal requires a fresh current certificate with $M_{i,j} \ge 2$. Components never combine and duplicate/older cycles are rejected. There is no round-30/60 switch, removal quota or arbitrary certificate age limit. Long retention nevertheless risks accumulating correlated false evidence.

Exact current zero delta ($r_i = 0$), or a fingerprint match to the immediately previous model sent to that client, is a separate sendback observation. Sendback on both selected traps in consecutive complete cycles can remove. Near-zero norms alone are not exact equality, and warm-up cannot bypass this requirement.

F. Aggregation and Recovery

Trap responses never aggregate. Certified S starts a two-cycle bounded review; fresh strong evidence can renew it, but stored memory and weak C evidence cannot. Expired unconfirmed quarantine is released. Two clean cycles rehabilitate, followed by one extra clean probation cycle. Weak candidates can aggregate on ordinary rounds. New quarantine/removal also excludes the decision-round update. Eligible deltas are averaged equally, not weighted by private sample counts. An empty eligible set leaves the global model unchanged.

Independent streams seed partitioning, attacker selection, initialization, training, grouping, traps and attack randomness. Checkpoints retain RNG/loaders, attacker histories, selected probes, review state and memory. Resume rejects changed protocol/source hashes. Recovery does not create detector evidence.

Algorithm 1 Component-specific fresh-certificate confirmation

Require: complete A/B measurements and a distinct cycle
compute unchanged v15.3 nominations and strict certificates
for all active clients i **do**
  if fresh statistical certificate for component j **then**
    $M_{i,j} \leftarrow M_{i,j} + 1$; interrupt clean streak
    if $M_{i,j} \geq 2$ **then**
      remove i
    end if
  else if complete R cycle **then**
    advance consecutive clean-cycle streak
    if two clean cycles **then**
      clear all $M_{i,j}$
    end if
  else
    retain counts; interrupt clean streak
  end if
  apply separate consecutive-cycle exact-sendback rule
end for
release unconfirmed quarantine when bounded review expires

IV. Experimental Design

The intended grid is CIFAR-10, IID/non-IID, FR1–FR4, 30%/40% free riders: sixteen 100-round executions with master seed 42. IID uses shuffled equal-size partitions. Non-IID uses class-wise Dirichlet allocation with $\alpha = 0.5$, retrying to obtain nonempty client datasets. Honest clients use three local SGD epochs, batch size 32, learning rate 0.01 and momentum 0.9. Images use fixed channel normalization without augmentation. The CNN has three convolution/ BatchNorm blocks (32, 64, 128 channels), pooling, a 256-unit FC layer and ten logits.

Evaluation reports TP/FP/FN for permanent removal, precision, recall, F1 and honest false-removal rate. Precision is undefined when nobody is removed. Complete-detection round t_{all} exists only if all free riders are removed; otherwise report NR, not the last observed removal. Utility, final honest candidates, unique honest clients quarantined, quarantine client-rounds and whole-process resources are separate. Reused identities are client-cases, not independent statistical samples. A single seed does not support sample-SD or cross-seed confidence intervals.

The reporting pipeline requires all sixteen completion markers, 100 rounds per run, unique labelled removals, fixed evaluation truth, complete initial A/B coverage, consistent active-client logs and confusion/exclusion summaries, and matching frozen detector provenance. It neither resumes training nor selects favorable runs. Incomplete data stay blank; completed detection failures remain. Earlier MNIST/v8/v15.3 outcomes are not substituted.

V. Results Templates

Across sixteen completed seed-42 executions, 560/560 free-rider cases and 0/1040 honest-client cases were permanently removed. Pooled removal recall is 100.00%, precision is 100.00%, and F1 is 100.00%. There are 10 honest-client cases ever quarantined and 182 final honest-candidate cases. These are repeated simulation cases, not independent clients or held-out seeds. See the per-condition tables; incomplete removal remains a failure rather than a missing result.

TABLE I

REMOVAL PERFORMANCE. NR MEANS COMPLETE REMOVAL WAS NOT ACHIEVED BY ROUND 100; – DENOTES PENDING/UNDEFINED EVIDENCE.

FR (%)	Split	Attack	TP	FP	FN	Recall (%)	Precision (%)	F1 (%)	t_{all}
30	IID	FR1	30	0	0	100.00	100.00	100.00	30
30	IID	FR2	30	0	0	100.00	100.00	100.00	30
30	IID	FR3	30	0	0	100.00	100.00	100.00	30
30	IID	FR4	30	0	0	100.00	100.00	100.00	30
30	Non-IID	FR1	30	0	0	100.00	100.00	100.00	30
30	Non-IID	FR2	30	0	0	100.00	100.00	100.00	30
30	Non-IID	FR3	30	0	0	100.00	100.00	100.00	80
30	Non-IID	FR4	30	0	0	100.00	100.00	100.00	30
40	IID	FR1	40	0	0	100.00	100.00	100.00	30
40	IID	FR2	40	0	0	100.00	100.00	100.00	30
40	IID	FR3	40	0	0	100.00	100.00	100.00	30
40	IID	FR4	40	0	0	100.00	100.00	100.00	30
40	Non-IID	FR1	40	0	0	100.00	100.00	100.00	30
40	Non-IID	FR2	40	0	0	100.00	100.00	100.00	30
40	Non-IID	FR3	40	0	0	100.00	100.00	100.00	60
40	Non-IID	FR4	40	0	0	100.00	100.00	100.00	30

TABLE II

UTILITY AND HONEST-CLIENT SAFETY. C_H IS FINAL HONEST-CANDIDATE COUNT, Q_H UNIQUE HONEST CLIENTS EVER QUARANTINED, E_H HONEST QUARANTINE CLIENT-ROUNDS, AND Q_T ALL CLIENTS QUARANTINED AT COMPLETION.

FR (%)	Split	Attack	Accuracy (%)	C_H	Q_H	E_H	Q_T
30	IID	FR1	78.07	2	0	0	0
30	IID	FR2	77.93	0	0	0	0
30	IID	FR3	78.17	2	0	0	0
30	IID	FR4	78.03	7	0	0	0
30	Non-IID	FR1	75.80	8	0	0	0
30	Non-IID	FR2	75.58	2	0	0	0
30	Non-IID	FR3	75.36	70	1	20	0
30	Non-IID	FR4	75.78	7	0	0	0
40	IID	FR1	77.46	0	0	0	0
40	IID	FR2	77.65	0	0	0	0
40	IID	FR3	77.27	4	0	0	0
40	IID	FR4	77.36	8	0	0	0
40	Non-IID	FR1	75.28	9	0	0	0
40	Non-IID	FR2	75.43	4	0	0	0
40	Non-IID	FR3	74.89	52	9	180	0
40	Non-IID	FR4	75.14	7	0	0	0

TABLE III

WHOLE-SIMULATOR RESOURCES, NOT ISOLATED DETECTOR OVERHEAD. MEAN LOGGED ROUND TIME EXCLUDES SUBSEQUENT CHECKPOINT/LOG PUBLICATION; WALL TIME INCLUDES INITIALIZATION AND PUBLICATION. RSS INCLUDES TRAINING, AGGREGATION AND SIMULATED ATTACKER HISTORIES.

FR (%)	Split	Attack	Mean round (s)	Peak RSS (MiB)	Wall time (min)
30	IID	FR1	88.18	987.07	152.4
30	IID	FR2	97.78	780.39	165.6
30	IID	FR3	99.87	1796.55	179.0
30	IID	FR4	119.07	1697.04	217.4
30	Non-IID	FR1	95.03	983.02	162.9
30	Non-IID	FR2	89.56	752.98	151.7
30	Non-IID	FR3	108.38	1807.34	193.5
30	Non-IID	FR4	95.61	1791.95	173.6
40	IID	FR1	80.97	1013.29	139.7
40	IID	FR2	77.38	734.54	131.3
40	IID	FR3	104.21	1936.14	198.0
40	IID	FR4	100.07	1977.06	193.8
40	Non-IID	FR1	78.42	1040.16	135.9
40	Non-IID	FR2	75.80	794.95	128.9
40	Non-IID	FR3	84.85	2180.93	160.1
40	Non-IID	FR4	78.79	2118.29	149.5

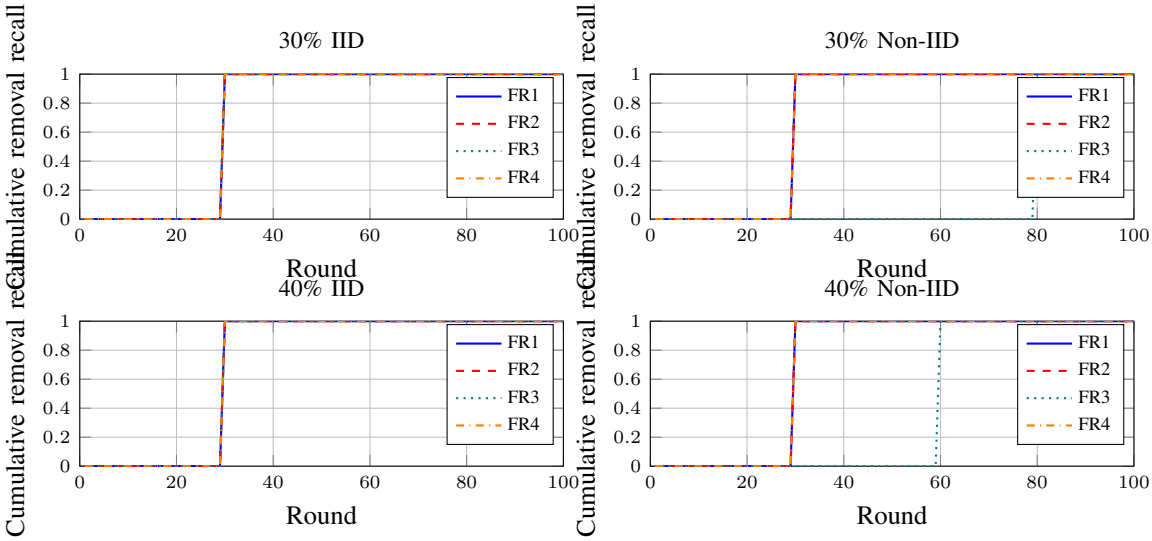


Fig. 1. Cumulative removal recall, seed 42. All attacks are retained, including misses; coincident curves may overlap.

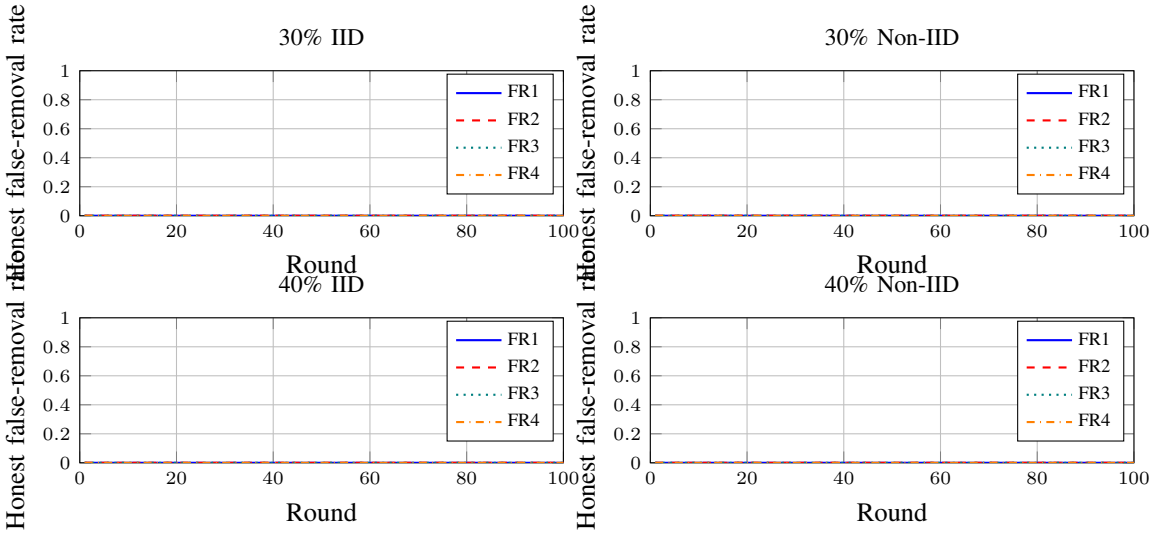


Fig. 2. Honest false-removal rate, seed 42. All attacks are retained, including misses; coincident curves may overlap.

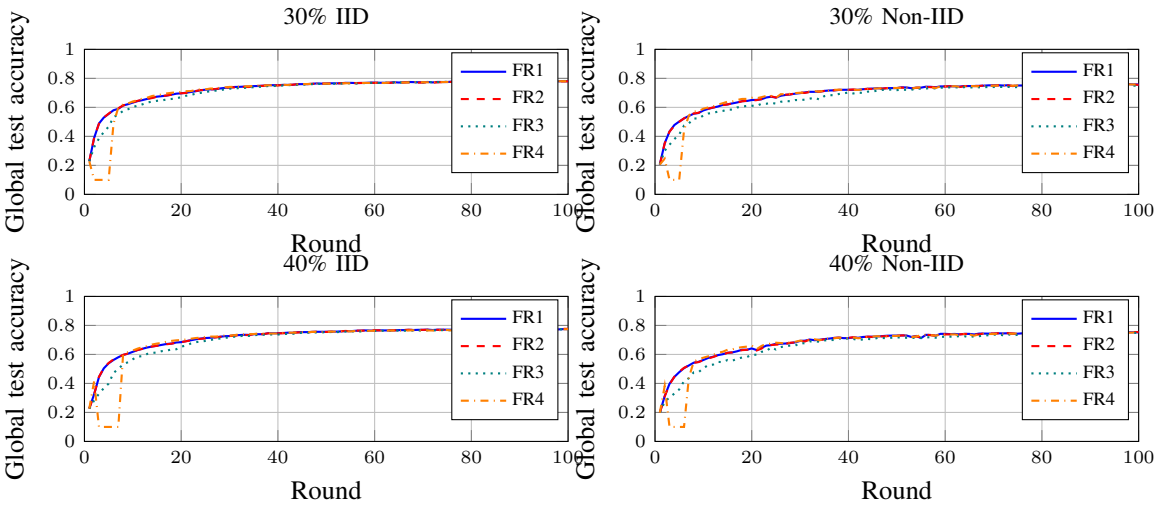


Fig. 3. Global test accuracy, seed 42. All attacks are retained, including misses; coincident curves may overlap.

VI. Cost and Limitations

Detector features retain $O(NL)$ norm/profile summaries and a few counters per client, plus two trap models. This is not total simulator memory: training, aggregation and simulated FR3/FR4 histories retain additional state. Extraction reads model coordinates; existing leave-one-out peer comparisons are roughly quadratic in client count, and bootstrap scratch space is $O(BN)$ for $B = 1000$.

Whole-process RSS and round duration cannot isolate detector overhead. A dedicated detector-only benchmark, communication accounting and matched no-detector controls remain required. "Lightweight" denotes compact statistics without auxiliary detector training, not a deployment-scale bound.

The policy was developed using earlier seed-42 failures. Changing prevalence to 30% is not held-out-seed validation. Neither zero observed honest removals nor a stable bootstrap split guarantees future error rates. Honest clients can pass certificates, including the retained profile-density gate. Certificates share client data and trajectories and are not statistically independent. Minimum cohort support can miss isolated free riders; long-lived evidence can eventually remove honest clients unless clearance occurs.

Required follow-up includes new seeds, honest-only controls, longer horizons, different history/noise parameters, mixed/adaptive attacks, partial participation and other architectures. Matched ablations must distinguish nomination, certificate and memory effects. False candidate flags and temporary exclusions must remain visible even when permanent-removal precision is high.

VII. Conclusion

This revision specifies a strict attack-label-blind gate and component-specific memory for intermittent fresh evidence. It supplies blank reproducible templates rather than predicting a desired detection count. Populated findings describe only the completed grid and require review alongside temporary honest exclusions and single-seed limitations.

References

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